

Algebraic Topology: a beginner's course

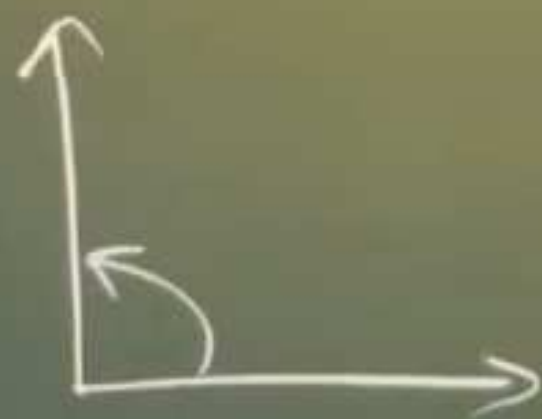
This lecture:

**AlgTop11: Rational curvature,
winding and turning**

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Lecture 11: Rational curvature, winding & turning





$$\underline{A2} \quad \theta = 90^\circ$$

$$\underline{A3} \quad \theta = 100 \text{ grad.}$$

A4 (much more natural
& better)

$$\boxed{\tau = \frac{1}{4}} \quad \checkmark \checkmark$$

Q How to describe the amount
of turning?

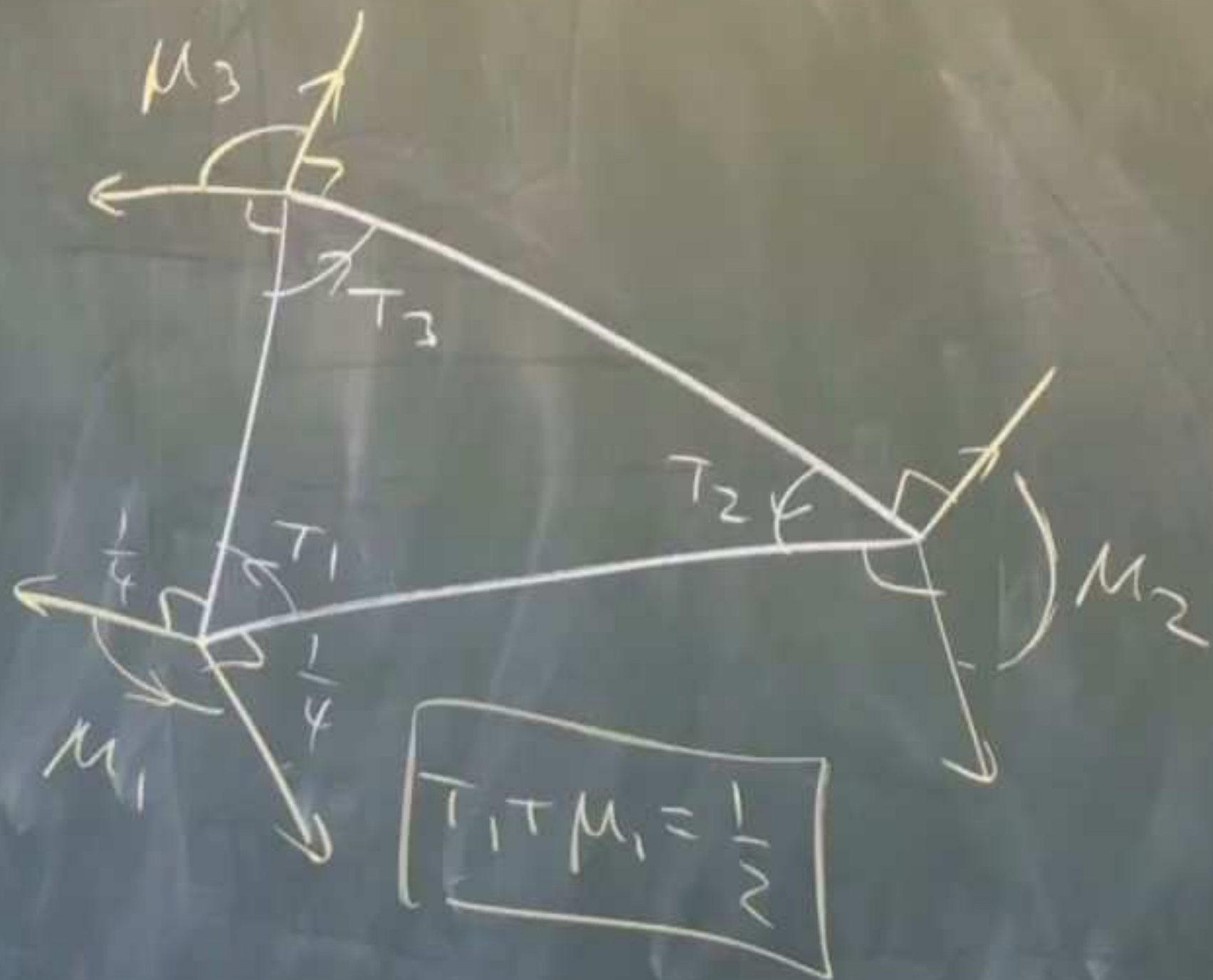
$$\underline{A1} \quad \theta = \frac{\pi}{2} = \frac{3.1415 \dots}{2} = 1.5 \dots$$

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next in the lecture:

**AlgTop11c: Rational curvature,
winding and turning (cont.)**

Alternate approach

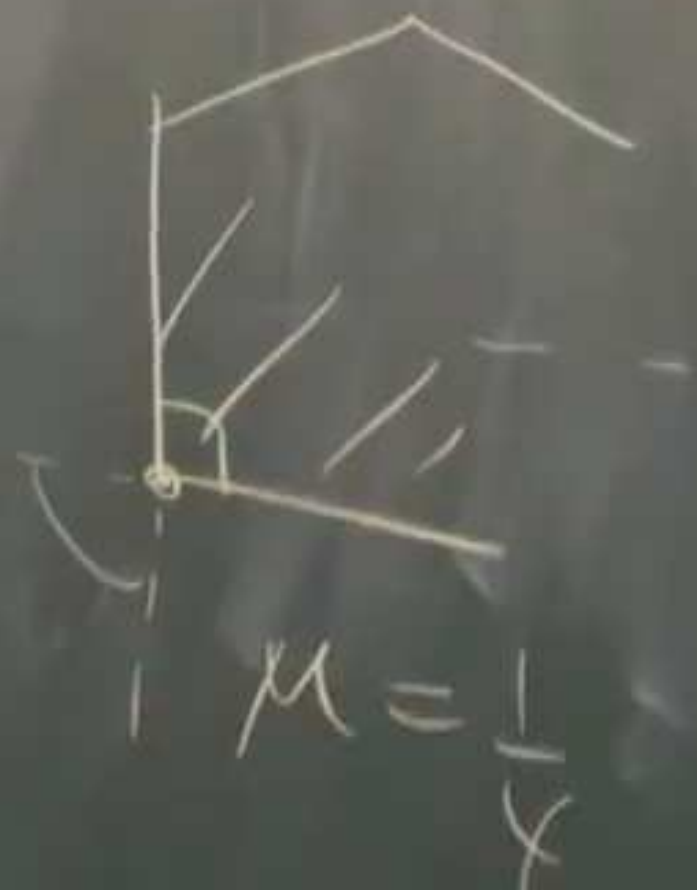
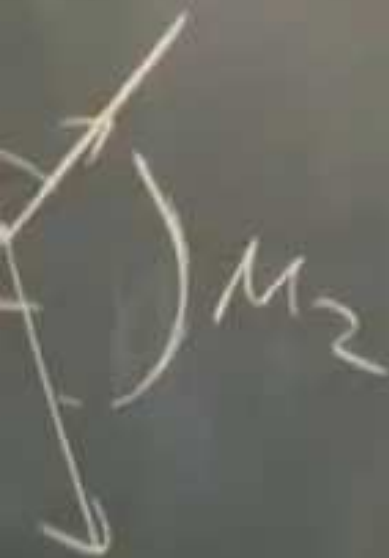


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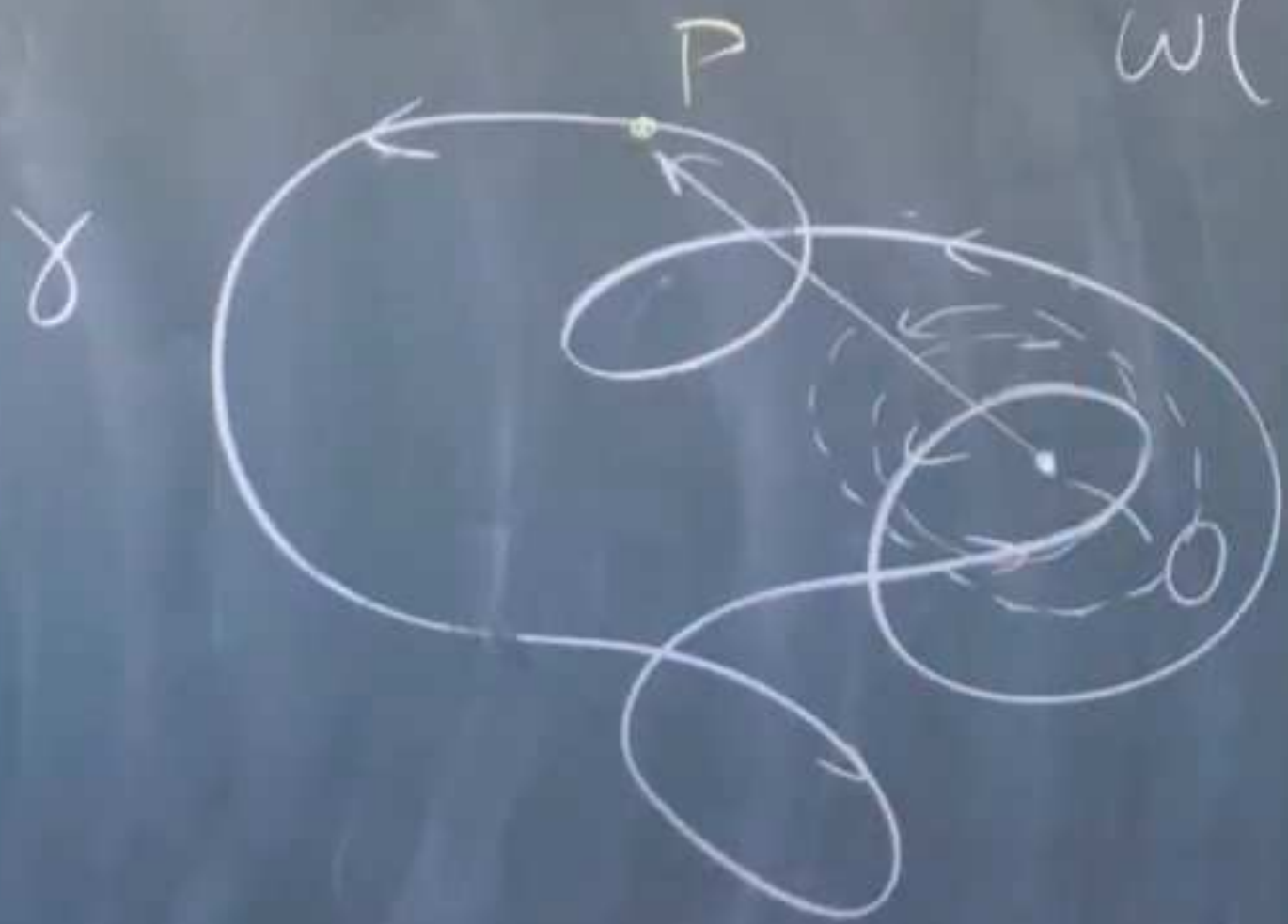
Define $\mu_i = \overset{\text{rational}}{\underbrace{\text{curvature of } \Delta \text{ at } A_i}}$

Total curvature of convex polygon

$$= \sum \mu_i = 1$$



Winding numbers of a curve



$w(\gamma; O)$ = the total
amount of turning
of vector $\vec{OP}$, $P \in \gamma$

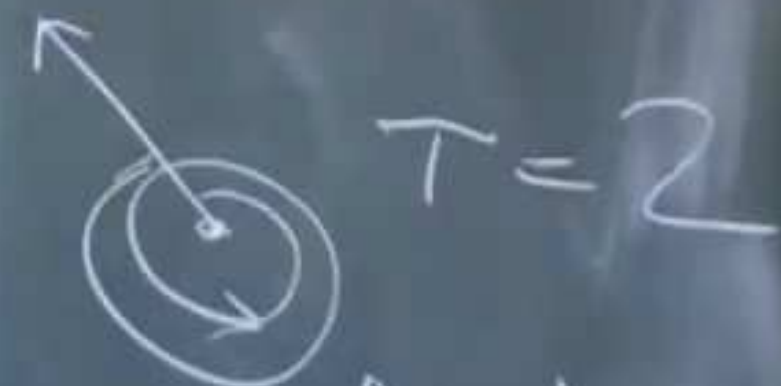


$$T=2$$

Winding numbers of a curve



$w(\gamma; O)$ = the total amount of turning of vector $\vec{OP}$, $P \in \gamma$



Sum of turns of individual segments of curve.